

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-051***

South Greenwich-Willard 69 kV

March 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: (1) Direct Connections, which are new facilities and/or facilities upgrades needed to connect the generator to the PJM network, and (2) Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances a generator interconnection may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

6011 Greenwich Windpark LLC (Greenwich Windpark) proposes to install PJM Project #AC1-051, a 60.0 MW (7.8 MW capacity) wind generating facility. The proposed Point of Interconnection is on the Greenwich-South Greenwich section of AEP's South Greenwich-Willard 69 kV circuit (see Figure 1). The proposed generating facility is located in the vicinity of Greenwich, Ohio (see Figure 2).

The requested in service date is December 31, 2019.

Attachment Facilities

Point of Interconnection (South Greenwich-Willard 69 kV)

The proposed generating facility is to be tapped between the Greenwich and the South Greenwich stations on the South Greenwich-Willard 69 kV circuit. The generating facility will be separated from the tapping point by a circuit breaker. In addition, the Willard 69 kV station configuration must be modified to operate as a ring bus. A transfer trip relaying scheme will be required to open the wind farm breaker when the breakers at the Willard station open the Willard end of the Willard – South Greenwich 69 kV line (see Figure 1).

The existing Willard 69 kV station does not have enough space to accommodate the required reconfiguration. To achieve the required space, the Willard 69 kV station will need to be expanded to a dimension of 200 ft. by 200 ft. To attain the required space for the Willard 69 kV station expansion, a 3 acre plot of land will need to be added in the Northwest direction of the existing Willard 69 kV station. AEP to obtain the additional land required to expand the existing Willard 69 kV station and the costs will be reimbursable by Greenwich Windpark.

In addition, the proposed AC1-051 Wind generating facility will likely be required to be temporarily removed from service in the event of any facility outage (planned or unplanned) that leaves either of the 69 kV paths from Willard to Carrothers or Willard to Howard open, thereby leaving the Willard-S Greenwich 69 kV line part of an extended radial path (See Figure 1). To ensure that this removal occurs with minimal delay, AEP Operations will have Supervisory trip control for the AC1-051 project breaker.

The point of interconnection is also to include 69 kV metering, SCADA, and associated equipment. Protection relays in the surrounding area will need to be reset to accommodate the addition of the new project.

New Switching Station Work:

- Construct a new 69 kV switching station with one (1) 69 kV circuit breaker. Installation of associated protection and control equipment, 69 kV line risers, SCADA (see Figure 1).
- **Estimated Station Cost: \$2,000,000**

Direct Connection Cost Estimate

The total preliminary cost estimate for Direct Connection work is given in the following tables below.

For AEP building Direct Connection cost estimates:

Description	Total Cost
Greenwich-South Greenwich 69 kV T-Line Cut In	\$750,000
Willard-Greenwich 69 kV T-Line Work	\$250,000
Carrothers-Willard 69 kV T-Line Work	\$250,000
Total	\$1,250,000

Table 1

Non-Direct Connection Cost Estimate

The total preliminary cost estimate for Non-Direct Connection work is given in the following tables below:

For AEP building Direct Connection cost estimates:

Description	Estimated Cost
69 kV Revenue Metering	\$150,000
Expand Willard 69 kV Substation	\$1,000,000
Total	\$1,150,000

Table 2

Interconnection Customer Requirements

It is understood that Greenwich Windpark is responsible for all costs associated with this interconnection. The costs above are reimbursable to AEP. The cost of Greenwich Windpark's generating plant and the costs for the line connecting the generating plant to Greenwich Windpark's switching station are not included in this report; these are assumed to be Greenwich Windpark's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for American Electric Power to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

Requirements from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

AEP Requirements

The Interconnection Customer will be required to comply with all AEP Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "Requirements for Connection of New Facilities or Changes to Existing Facilities Connected to the AEP Transmission System" document located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-aep/aep-interconnection-requirements.ashx>

Network Impacts

The Queue Project AC1-051 was evaluated as a 60.0 MW (Capacity 7.8 MW) injection into the N. Greenwich 69 kV substation (tap along the Willard – S. Greenwich 69 kV line) in the AEP area. Project AC1-051 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC1-051 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Base Case Used

Summer Peak Analysis – 2020 Case

Contingency Descriptions

The following contingencies resulted in overloads:

None

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None

Short Circuit

(Summary of impacted circuit breakers)

New circuit breakers found to be over-duty:

None

Affected System Analysis & Mitigation

LGEE Impacts:

None

MISO Impacts:

None

Duke, Progress & TVA Impacts:

None

OVEC Impacts:

None

Stability and Reactive Power Requirement

(Results of the dynamic studies should be inserted here)

Stability Results will be confirmed in the Facilities Study phase of the project.

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

Note: There is actually a more severe N-1 condition which was (probably) not studied. Loss of the S. Greenwich load should be considered a valid N-1 condition, simulating outage of either the 69/12 kV transformer (which could be cleared by operation of the 69 kV circuit switcher) or operation of the breaker on the single 12 kV feeder. This contingency exposure will increase the risk of curtailment.

Contingency			Affected Area	Facility Description	Bus		Loading		Rating		MW Con.		
#	Type	Name			From	To	Cir.	PF	Initial	Final		Type	MVA
1	Non	Non	AEP - AEP	05GREENWIC-05BOUGHTN8 69 kV line	245662	245651	1	DC	17.6	102.4	NR	50	60
2	Non	Non	AEP - AEP	AC1-051 TAP-05GREENWIC 69 kV line	925750	245662	1	DC	11.4	108.6	NR	50	60

Table 3

Light Load Analysis

No light load thermal impacts

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None

Schedule

It is anticipated that the time between receipt of executed agreements and Commercial Operation may range from 12 to 18 months if no line work is required. If line work is required, construction time would be between 24 to 36 months after signing an interconnection agreement.

Note: The time provided between anticipated normal completion of System Impact, Facilities Studies, subsequent execution of ISA and ICSA documents, and the proposed Backfeed Date is shorter than usual and may be difficult to achieve.

Conclusion

Based upon the results of this System Impact Study, the construction of the 60.0 MW (7.8 MW Capacity) wind generating facility of Greenwich Windpark (PJM Project #AC1-051) will require the following additional interconnection charges. This plan of service will interconnect the proposed wind generating facility in a manner that will provide operational reliability and flexibility to both the AEP system and the Greenwich Windpark generating facility.

Cost Breakdown for Point of Interconnection (South Greenwich-Willard 69 kV)			
Type of Network Upgrade	Network Upgrade Number	Description	Estimated Cost
Direct Connection	n5567	New 69 kV Switching Station	\$2,000,000
Non-Direct Connection	n5568	Greenwich-South Greenwich 69 kV T-Line Cut In	\$750,000
	n5569	Willard-Greenwich 69 kV T-Line Work	\$250,000
	n5570	Carrothers-Willard 69 kV T-Line Work	\$250,000

Cost Breakdown for Point of Interconnection (South Greenwich-Willard 69 kV)			
Type of Network Upgrade	Network Upgrade Number	Description	Estimated Cost
	n5571	69 kV Revenue Metering	\$150,000
	n5572	Expand Willard 69 kV Substation	\$1,000,000
Total Estimated Cost for Project AC1-051			\$4,400,000

Table 4

The estimates are preliminary in nature, as they were determined without the benefit of detailed engineering studies. Final estimates will require an on-site review and coordination to determine final construction requirements.

Figure 1: Point of Interconnection (South Greenwich-Willard 69 kV)
Single-Line Diagram

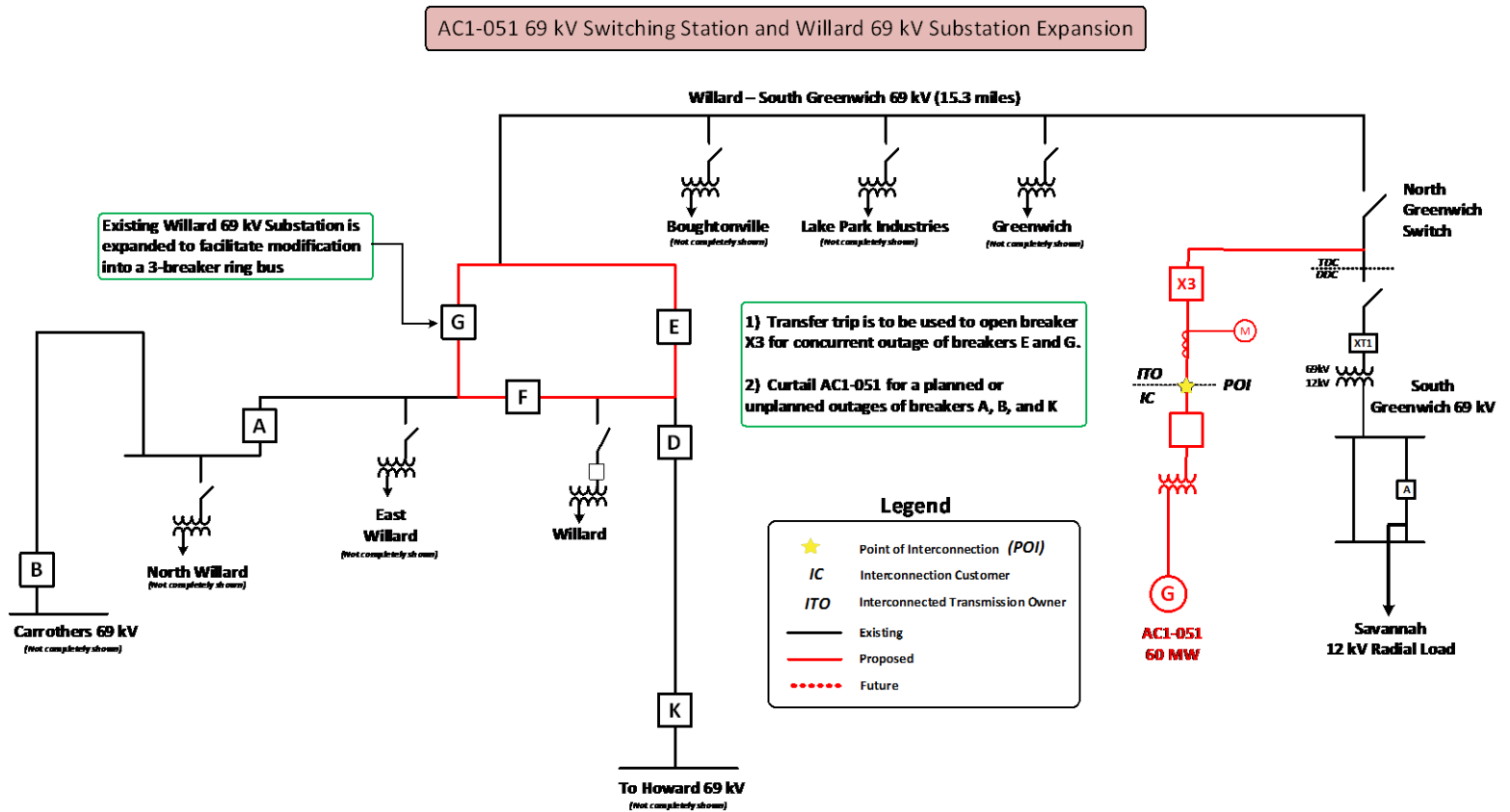


Figure 2: Point of Interconnection (South Greenwich-Willard 69 kV)

